

Sudarshan Babu

Research Scientist – Generative Modeling

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Research Summary

Research scientist specializing in diffusion and score-based generative models for image, video, and 3D scene generation, extending to cryo-EM volumes, 3D molecular structures, and set-structured single-cell data. Builds data-efficient generative systems with diffusion, flow matching, hypernetworks, and implicit neural representations, grounded in out-of-distribution generalization.

Core Expertise

Research Diffusion models, 3D generation, small molecule design, single cell modelling
Methods Hypernetworks, implicit neural representations (INRs), transformers, meta-learning

Professional Experience

Sep 2025–Present **Research Scientist**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL
Developed scLDM, a latent diffusion model for single-cell data generation
Built implicit neural field model for cryo-ET volume generation

Jan 2025–Sep 2025 **AI Fellow**, *Chan Zuckerberg Biohub Chicago*, Chicago, IL
Developed HyperDiffusionFields (HyDiF), combining diffusion models with hypernetworks for implicit molecular neural fields generation

Oct 2021–Jan 2022 **Research Scientist Intern**, *Amazon*, Bellevue, WA
Designed curriculum learning strategies for efficient neural network training

Feb 2021–Jun 2021 **Research Scientist Intern**, *NVIDIA*, Remote
Built systems for training models on non-stationary long-tail distributions
Contributed to 4 patent applications in adaptive learning and distributed training systems

Education

Oct 2019–2025 **Ph.D., Computer Science**, *Toyota Technological Institute at University of Chicago*, Chicago, IL, GPA: 3.90/4.0
Advisors: Prof. Michael Maire, Prof. Greg Shakhnarovich
Dissertation: Acquiring and Adapting Priors for Novel Tasks via Neural Meta-Architectures

Publications

Under submission **Atelier: Learning Local Self-Supervised Features for CryoEM Volumes via Hypernetworks**
Phillip Lo*, Sudarshan Babu*, Dari Kimanius[†], Aly A. Khan[†]

ICML '26 **Virtual Cells Need Context, Not Just Scale**
Payam Dibaeinia, Sudarshan Babu, Mei Knudson, Ali ElSheikh, et al.

ICML '26 **A Scalable Latent Diffusion Model for Single-Cell Gene Expression Data**
Giovanni Palla, Sudarshan Babu, Payam Dibaeinia, et al.

NeurIPS WS '25 **HyperDiffusionFields (HyDiF): Diffusion-Guided Hypernetworks for Learning Implicit Molecular Neural Fields**
Sudarshan Babu*, Phillip Lo*, Xiao Zhang, et al.

ICML '24 **HyperFields: Towards Zero-Shot Generation of NeRFs from Text**
Sudarshan Babu*, Richard Liu*, Avery Zhou*, et al.

SIGGRAPH Asia '24 **iSeg: Interactive 3D Segmentation via Interactive Attention**
Itai Lang*, Fei Xu*, Dale Decatur, Sudarshan Babu, Rana Hanocka

NeurIPS '21 **Online Meta-Learning via Learning with Layer-Distributed Memory**
Sudarshan Babu, Pedro Savarese, Michael Maire

CVPR '21 **Domain-Independent Dominance of Adaptive Methods**
Pedro Savarese, David McAllester, Sudarshan Babu, Michael Maire

Preprint '20 **HyperNetwork Designs for Improved Classification and Robust Meta-Learning**
Sudarshan Babu, Pedro Savarese, Michael Maire

* equal contribution † co-senior authorship

Service

Reviewing CVPR, ICCV, ECCV, NeurIPS, ICLR, SIGGRAPH, RECOMB

Teaching Teaching Assistant for Machine Learning, Prof. Greg Shakhnarovich, Fall 2020

Organization Co-organizer: Computational Biology Workshop @ TTIC, Sep 2024; Annual TTI-C Student Workshop, Feb 2020; Computer Vision Reading Group, Oct 2020 – Feb 2024